Masking Empirical Mode Decomposition-Based Hybrid Features for Recognition of Motor Imagery in EEG

Wanzhong Chen*, Yang You
College of Communication Engineering
Jilin University
Changchun, China
e-mail: chenwz@jlu.edu.cn, youyang15@mails.jlu.edu.cn

Abstract—**Brain computer interface (BCI) in electroencephalogram (EEG) is of great important but it is challenging to manage the nonstationary EEG. In this paper, a newly-developed method, namely masking empirical mode decomposition (MEMD), is applied to deal with motor imagery (MI) recognition tasks. The windowed EEG is decomposed by MEMD and hybrid features are then extracted from the first two intrinsic mode functions (IMFs). Kruskal-Wallis test is carried out to extract significant features and they are finally fed into linear discriminant analysis (LDA) for classification. Results show our proposed approach can achieve the highest accuracy of 88.35% as well as the maximal mutual information of 0.6846. The presented technique is comparable or superior to other methods and is proven useful.**

Keywords-brain-computer interface (BCI); electroencephalogram (EEG); masking empirical mode decomposition (MEMD); motor imagery (MI)

I. Introduction

Brain-computer interface (BCI) aims to provide a direct communication pathway between brain and external devices [1]. In recent years, with the development of BCI, this new human-computer interaction method has attracted people's attention for its potential applications in rehabilitation engineering, intelligent assistive robot and etc. [2]. In motor imagery (MI), when people imagining an action without execution, the corresponding brain sensorimotor areas are activated and the same electroencephalogram (EEG) generates, as if the action is done [3]. During mental imagination of different motor tasks, event-related de-synchronization and synchronization (ERD/ERS) phenomena, which induce the change of cerebral cortex potential, exhibit in the EEG [4]. Imagining left or right hand movement, an obvious ERD phenomenon can be observed from the mu (8~12Hz) and beta (13–30 Hz) rhythms of EEG which gathered from the contralateral sensorimotor area of cerebral cortex [5]. These characters lay the foundation of MI-based BCI.

Feature extraction of EEG is the key procedure of MI recognition and the existing feature extraction methods mainly include: power spectral density (PSD) [6], adaptive auto-regressive (AAR) [7], energy-based wavelet transform (WT) [8], wavelet packet transform (WPT) [9], empirical mode decomposition (EMD) [10] and so on. For example, multi-feature EEG classification method with using wavelet-based fuzzy approximate entropy introduced in [7], can improve the accuracy and achieve outstanding results. The amplitude–frequency analysis to extract the phase space features which contained the nonlinear information proposed in [11] is confirmed to improve the classification results in the BCI. Fuzzy support vector machine (FSVM) for classifying motor imagery tasks with wavelet-based features reported in [8] could reduce the effects of noise or outliers in the online systems. Considering the non-stationary nature of EEG signal, empirical mode decomposition (EMD) is also applied for EEG analysis in different applications [10]. Recently in [10], the potential of EMD analysis in MI data processing is discussed with a note that relevant features must be carefully selected for getting better accuracy. However, EMD hardly avoids the mode mixing and edge effect phenomenon, which makes the sub-signals being a lower signal-to-noise ratio. Despite EEMD is developed to improve the defects, the concomitant high time consuming is unacceptable in some situations [12]. The main motivation behind this paper is to develop a new approach that is capable of finding the optimal method for solve the mode mixing and edge effect phenomenon and also provide good classification performance.

In masking empirical mode decomposition (MEMD) [13], a known frequency masking signal is added to and subtracted from the original signal, and followed by performing EMD on these two processed signals, respectively. Under the circumstances, MEMD is able to effectively restrain the influence of mode mixing. Taking advantage of MEMD, a novel scheme of the left-right MI recognition in EEG is proposed in this research. Initially, the windowed EEG is filtered and MEMD is then employed to decompose the filtered EEG. Subsequently, the first two pairs of intrinsic mode functions (IMFs), corresponding to the *mu* and *beta* rhythms, are selected as useful sub-components. Energy, auto-regressive (AR) coefficients, morphological characteristics and fuzzy approximate entropy (*fApEn*) are extracted from the selected sub-components. After that, Kruskal-Wallis test is conducted on each feature and those with p-value lower than 0.05 are reserved and finally fed into a linear discriminant analysis (LDA) classifier for classification.

II. Material and Methods

A. EEG Data

In the current work, the BCI competition 2003 data set (motor imagery III) provided by Technical University of

978-1-5386-0484-7/17/$31.00 ©2017 IEEE

Graz is used to verify the effectiveness of our presented method. This data was recorded when a healthy volunteer was sitting in a chair with arm rests and was trying to control a feedback bar with imagery movements of left or right hands. The order of left and right cues was randomly determined. For detailed information please refer to [14].

B. MEMD

EMD is capable of decomposing an arbitrary real signal into a series of IMFs and each IMF is a frequency modulation and amplitude modulated (AM-FM) sequence. For signals with intermittent oscillations, one intrinsic mode obtained by EMD can comprise oscillations with a variety of wavelengths at different temporal locations. The simultaneous appearance of these disparate oscillations is known as the mode mixing issue which results in complicate analysis and obscure physical meanings [15]. MEMD was proposed to soften the influence of mode mixing.

The main steps of MEMD are summarized as follows [13]:

Suppose the original signal $x(n)=\{x(1),x(2),...,x(N)\}$. Firstly, EMD is inflicted on $x(n)$ and its IMFs are obtained.

Secondly, Hilbert transform is applied to the first IMF to extract its instantaneous amplitude a_{IMF1} and instantaneous frequency f_{IMF1} . Accordingly, the instantaneous amplitude a_z and instantaneous frequency f_z of the frequency masking signal are given by:

$$a_z=\frac{1.6}{N}\sum_{i=1}^{N}a_{IMF1}(i)\ , \tag{1}$$

$$f_z=\sum_{i=1}^{N}a_{IMF1}(i)f_{IMF1}^2(i)\Big/\sum_{i=1}^{N}a_{IMF1}(i)f_{IMF1}(i)\ . \tag{2}$$

Thirdly, inserting the masking signal into $x(n)$ and two new sequences $x^+(n)$ and $x^-(n)$ are obtained as:

$$x^+(n)=x(n)+z(n)\ , \tag{3}$$

$$x^-(n)=x(n)-z(n)\ . \tag{4}$$

EMD is performed on $x^+(n)$ and $x^-(n)$ to obtain the IMFs $y^+(n)$ of $x^+(n)$ and $y^-(n)$ of $x^-(n)$, respectively.

Finally, the IMFs $y(n)$ of $x(n)$ using MEMD are calculated as: $y(n)=(y^+(n)+y^-(n))/2$.

C. Feature Extraction

In our investigation, four kinds of features including energy, AR coefficients, morphological characteristics and fApEn are utilized. It should be noted that only the first two IMFs are considered as the fact these two IMFs contain the main rhythms of MI.

For a given sequence $h(n)=\{h(1),h(2),...,x(N)\}$, these aforementioned features are described below.

1) Energy

Energy is a crucial parameter for the identification of left-right MI EEG.The expression is given,

$$E=\sum_{n=0}^{N\text{-}1}\left|H_t(n)\right|^2 \tag{5}$$

where N is the length of the signal.

2) AR coefficients

AR models are tested to be a kind of effective feature in MI recognition tasks and have been widely used in BCI research [11]. The AR coefficients embody the time-varying features of signals [17]. At the same time, AR is good at modeling the EEG as filtered white noise with certain preferred energy bands, which makes it fits for EEG signals analysis especially for EEG signals. Given a signal $X(t)$, the estimated equation in time domain is [11],

$$X(t)=\sum_{i=1}^{p}a(i)X(t-i)+e(t) \tag{6}$$

where $e(t)$ is the added white noise. In this research, the AR model is estimated with Yule-Walker method. The features from first level component $H_t(n)$ produced by MEMD are extracted. Through sufficient experiments test, the proper AR model orders are fixed to seven.

3) Morphological features

Five morphological features are calculated which were already described by Kalatzis [19]. Denote $X(t)$ as the considered EEG signal, its employed morphological features are as follows [19],

$$\begin{cases} AA=\max\left|x(t)\right| \\ PA=\sum_t 0.5*\left[x(t)+\left|x(t)\right|\right] \\ NA=\sum_t 0.5*\left[x(t)-\left|x(t)\right|\right] \\ TA=PA+NA \\ TAA=PA+\left|NA\right| \end{cases} \tag{7}$$

4) Fuzzy approximate entropy

The fApEn is obtained by the steps as follow [7]:

a) Given a signal $x(i)$ with the length of N, structure $\mathrm{X}_\mathrm{i}^\mathrm{m}$ vector.

$$x_i^m=\left\{x(i),x(i+1),\ldots,x(i+m-1)\right\}-x_0(i)$$

$$i = 1,2,\ldots,N-m+1, \tag{8}$$

where

$$x_0(i) = \frac{1}{m}\sum_{j=0}^{m-1} x(i+j) \tag{9}$$

b) Calculate the maximal distance d_{ij}^m between X_i^m and X_j^m and decide D_{ij}^m through a fuzzy membership function and the tolerance factor r as follow:

$$D_{ij}^m = \exp(-\frac{d_{ij}^2}{r}) \tag{10}$$

c) The *fApEn* is finally defined as:

$$fApEn(m,r,N) = \Phi^m(r) - \Phi^{m+1}(r) \tag{11}$$

and

$$\Phi^m(r) = (N-m+1)^{-1}\sum_{j=1,j\neq i}^{N-m+1} \ln C_r^m(i) \tag{12}$$

where

$$C_r^m(i) = (N-m+1)^{-1}\sum_{j=1,j\neq i}^{N-m+1} \ln C_r^m(i) \tag{13}$$

The MEMD decomposes signals into several components in different resolution. In this research, the fApEn is calculated from the first two high-frequency components of the decomposition. The multi-resolution fApEn will be obtained, which describes the complexity and regularity of data at multi-resolution.

III. Result

To validate the efficacy of our presented method, experiment is implemented in MATLAB R2016b environment with 4G RAM and AMD^{TM} 2.50GHz processor. In the experiment, the raw EEG is filtered by a band-pass filter with range of 0.5-30Hz. Then a sliding window with length of 383 samples is applied to the raw EEG. A typical EEG segment and its first two decomposed IMFs using MEMD are exhibited in Fig. 1.

As stated in prior section, feature extraction is conducted on the first IMFs and totally after MEMD is executed. Subsequently, Kruskal-Wallis test is adopted to estimate the discrimination ability of every feature. In this work, those significant features with lower p-value than 0.05 are selected and they constitute the final feature vector for classification. The p-values of extracted features are shown in Fig. 2. It can be observe from Fig. 2.

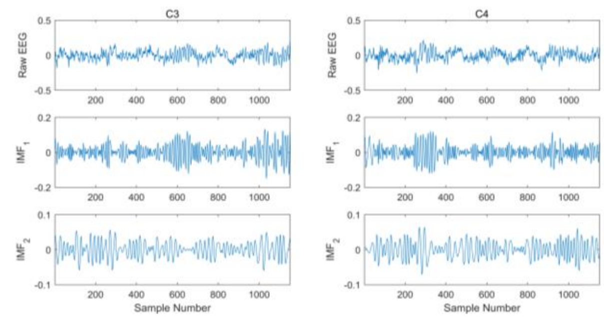

Figure 1. Typical EEG segment and its first two decomposed IMFs using MEMD.

To speed up the classification speed, the frequently-used but preeminent LDA classifier is employed. During classification phase, the training and testing sets are selected based on the explication of literature [20]. The classification accuracy for each segment is counted. Besides accuracy, MI is also an important evaluation criterion to measure the performance of MI-based BCI system. The values of classification accuracy and mutual information are illustrated in Fig. 3. As shown in Fig. 3, in initial period, both accuracy and mutual information increase as time goes from 3.5s to 7.5s.

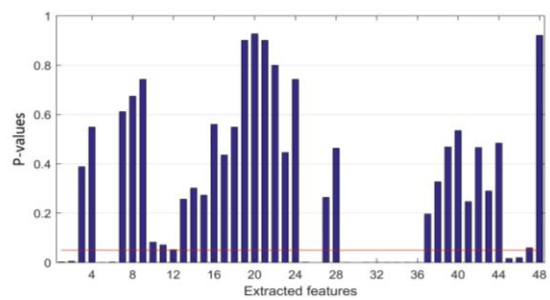

Figure 2. *P*-values of extracted features.

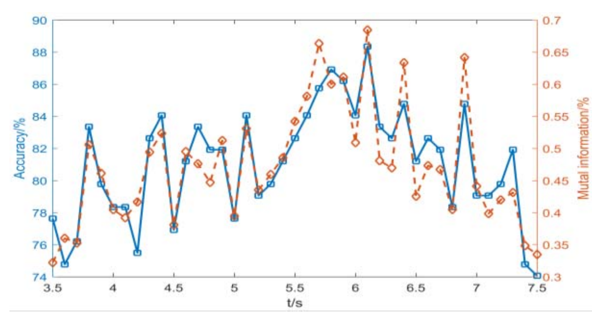

Figure 3. Values of accuracy and mutual information for proposed method.

Table I lists the comparison of maximal MI between our method and other works on the dataset. The maximal MI acquired by our method can reach 0.75. The result is higher than the BCI competition champions and the approaches of recent works. Compared with other existing MI EEG recognition approaches, the advantages of the proposed technique are: (1) a new data-driven method, *i.e.*, MEMD is applied to MI-based BCI system to overcome the mode mixing issue of EMD; (2) the nonlinear feature fApEn is introduced to reveal the complexity of IMFs of EEG signal; (3) The proposed method is applicable and effective with low computational complexity.

TABLE I. COMPARISON OF MAXIMAL MUTUAL INFORMATION (MUI) BETWEEN OUR WORK AND OTHER METHODS ON DATASET III, BCI COMPETITION II

Methods	ACC	Maximal MI
BCIC II 1st winner[18]	89.29	0.61
BCIC II 2nd winner[18]	84.29	0.46
BCIC II 3rd winner[18]	82.86	0.45
LDA with AFAPS [16]	90.71	0.64
Higher-order statistics features[17]	89.29	0.65
FSVM with wavelet-based features[8]	87.86	0.66
LDA with AFAPS+ARPS[11]	90.71	0.67
Our work	88.35	0.68

IV. CONCLUSION

In this study, the newly-developed MEMD is applied for left-right MI EEG recognition tasks. As an improved algorithm of EMD, MEMD can effectively smooth envelope, restrain mode mixing and reduce the impact of noise by operation of adding masking signals. Four kinds of effective features are extracted from the first two IMFs. Kruskal-Wallis test is used to extract significant features and LDA classifier is exploited for final classification. The highest accuracy is 88.35%, the corresponding MuI is 0.68.

However, the accuracy and MI are still not satisfactory. Further direction of our research may include but not limit to modification of MEMD and application of other signal processing methods for MI EEG recognition.

ACKNOWLEDGMENT

This work is supported by the Specialized Research Fund for the Doctoral Program of Higher Education of China (Grant No. 20100061110029), Project 2016092 Support by Graduate Innovation Fund of Jilin University and Natural Science Foundation for Science and Technology Development Plan of Jilin Province, China (Grant No. 20150101191JC).

REFERENCES

[1] N. Birbaumer, “Breaking the silence: brain-computer interfaces (BCI) for communication and motor control,” Psychophysiology, vol. 43, Nov. 2006, pp. 517-532, doi:10.1111/j.1469-8986.2006.00456.x.

[2] Wang Xingyu, Jin Jing, Zhang Yu, Wang Bei, “Brain control: human-computer integration control based on brain computer interface,” Acta Automatica Sinica, vol. 39, Mar. 2013, pp. 208-221, doi:10.3724/SP.J.1004.2013.00208.

[3] M. Jeannerod, “Mental imagery in the motor context,” Neuropsychologia, vol. 33, Nov. 1995, pp. 1419-1432, doi: 10.1016/0028-3932(95)00073-C.

[4] G. Pfurtscheller, “Functional brain imaging based on ERD/ERS,” Vision Research, vol. 41, May. 2001, pp. 1257-1260, doi: 10.1016/S0042-6989(00)00235-2.

[5] D. J. Mcfarland, L. A. Miner, T. M. Vaughan, J. R. Wolpaw, “Mu and Beta Rhythm topographies during motor imagery and actual movements,” Brain Topogr.vol. 12, Mar. 2000, pp. 177-186, doi: 10.1023/A:1023437823106.

[6] G. Rodríguez-Bermudez, P.J. García-Laencina, J. Roca-González, J. Roca-Dorda, “Efficient feature selection and linear discrimination of EEG signals,” Neurocomputing, vol. 1115, Sep. 2013, pp. 161–165, doi:10.1016/j.neucom.2013.01.001.

[7] G. Pfurtscheller, C. Neuper, A. Schlögl, K. Luger, “Separability of EEG signals recorded during right and left motor imagery using adaptive autoregressive parameters,” IEEE Transactions on Rehabilitation Engineering, vol. 6, Sep. 1998, pp. 316-325,doi:10.1109/86.712230.

[8] Q. Xu, H. Zhou, Y. J. Wang, J. Huang, “Fuzzy support vector machine for classification of EEG signals using wavelet-based features,” Medical Engineering & Physics, vol. 31, Sep. 2009, pp. 858-865, doi:10.1016/j.medengphy.2009.04.005.

[9] W. Y. Hsu, Y. N. Sun, “EEG-based motor imagery analysis using weighted wavelet transform features,” Journal of Neuroscience Methods, vol. 176, Jan. 2009, pp. 310-318, doi:10.1016/j.jneumeth.2008.09.014.

[10] N. E. Huang, Z. Shen , S. R. Long, M. C. Wu, H. H. Shih, Q. Zheng, N. C. Yen, C. C. Tung, H. H. Liu, “The empirical mode decomposition and the Hilbert spectrum for nonlinear and nonstationary time series analysis,” Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, vol. 454, Jan. 1998, pp. 903-995, doi: 10.1098/rspa.1998.0193.

[11] Y. Fang, M. Chen, and X. Zheng, "Extracting features from phase space of EEG signals in brain–computer interfaces," Neurocomputing, vol. 151, March. 2015, pp. 1477–1485, doi:10.1016/j.neucom.2014.10.038.

[12] Z. H. Wu, N. E. Huang, “Ensemble empirical mode decomposition: a noise-assisted data analysis method,” Advances in Adaptive Data Analysis, vol.1, Jan. 2009, pp. 1-41, doi: 10.1142/S1793536909000047.

[13] R. Deering, J.F. Kaiser, " The use of a masking signal to improve empirical mode decomposition,” IEEE International Conference on Acoustics, vol. 4, May. 2005, pp. 485-488, doi: 10.1109/ICASSP.2005.1416051.

[14] Available from: http://www.bbci.de/competition/ii/.

[15] S. Zhang, Y. Hu, P. Li, H. Bao, W. Jiang, L. Qian, “On the computational complexity of the empirical mode decomposition algorithm,” China Mechanical Engineering, vol. 26, Oct. 2015, pp. 2613-2618, doi:10.3969/j.issn.1004-132X.2015.19.010.

[16] M. Y Chen, Y. Fang, X. Zheng, “Phase space reconstruction for improving the classification of single trial EEG,” Biomedical Signal Processing And Control, vol. 11, May. 2014, pp. 10-16, doi: 10.1016/j.bspc.2014.02.002.

[17] S. M. Zhou, J. Q. Gan, F. Sepulveda, “Classifying mental tasks based on features of higher-order statistics from EEG signals in brain–computer interface,” Information Sciences, vol. 178, Mar.2008, pp. 1629-1640, doi:10.1016/j.ins.2007.11.012.

[18] Available from: http://bbci.de/competition/ii/results/index.html.

[19] I. Kalatzis, N. Piliouras, E. Ventouras, et al. “Design and implementation of an SVM-based computer classification system for discriminating depressive patients from healthy controls using the P600 component of ERP signals”, Computer Methods & Programs in Biomedicine, vol. 75,Jul. 2004, pp.11-22, doi:10.1016/j.cmpb.2003.09.003.

[20] M. Hariharan, C. Y. Fook, R. Sindhu, B. Ilias, S. Yaacob, “A comparative study of wavelet families for classification of wrist motions,” Computers & Electrical Engineering, vol. 38, Nov. 2012, pp. 1798-1807, doi:10.1016/j.compeleceng.2012.08.009.